

John Alfred Chen

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PROFILE

Penultimate-year Mechanical and Mechatronic Engineering student at UTS with hands-on experience in SOLIDWORKS (CSWA), AS 1100 drawing packs, revision control, tolerance-aware design, mechanical assembly, and build/test support. Strong interest in precision manufacturing, workshop problem solving, and high-quality production of small, complex mechanical parts.

EDUCATION

Bachelor of Engineering (Honours), Mechanical and Mechatronic Engineering

University of Technology Sydney: Expected completion: 2027 | WAM: 77

EXPERIENCE

Mechanical Design Intern, OpenTech Designs

May 2025 to present

- Automated CAD configuration and drawing-pack tasks to speed iteration and reduce drafting errors
- Created and documented 230+ controlled part configurations with compliant AS 1100 drawings
- Logged and tracked design changes to maintain traceability across updates and production stages
- Worked with engineers, technicians, and machinists to resolve mark-ups and build updates

Testbench Vehicle Engineer, UTS Motorsports Autonomous

February 2024 to August 2025

- Designed and integrated a scaled testbench chassis to validate entire autonomous sensor suite
- Designated hardpoints to ensure reliable mounting, alignment and serviceability using FEA
- Structured repeatable validation tests to document faults, corrections, and improvements

EXTRA-CURRICULAR ACTIVITIES

Open-Source Accessible Holonomic Wheel Project

September 2025 to December 2025

- Designed low-cost, tolerance-aware mecanum wheels to support younger students unable to access commercial holonomic products
- Implemented configurations to support variant generation across dependent components
- Verified components for FDM 3D printing with focus on affordability, repeatability and simplicity

Affordable Robotic Arm Redesign Initiative

July 2024 to November 2024

- Continued development of a student bionic arm project after formal program conclusion, focusing on a simplified and more affordable alternative architecture
- Reduced degrees of freedom to improve mechanical robustness, cost-efficiency and ease of replication

SKILLS

- SOLIDWORKS, Fusion 360, Inventor, AutoCAD (parts, assemblies, drawings to AS 1100 standards)
- FEA, motion studies, drawing interpretation, GD&T and tolerance stack, fit-up considerations
- Change control, drawing packs, and build/test documentation
- Basic soldering, sensor wiring, and electromechanical prototyping
- FDM 3D printing, print tuning, and repeatable part validation

REFEREES

Available upon request